

PRAKTIKUM BASIS DATA

MODUL 8



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
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JURNAL/TP

Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
 - a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
 - b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_name NOT NULL, email, salary, commission_pct, hire_date NOT NULL.
 - c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Link Rekaman Layar :

<https://youtu.be/m4WDtYzzkVM>

Code :

-- NOMOR 1

```
create table department8 (  
    department_id char(4) primary key,  
    departmrnt_name varchar2(50),  
    manager_id char(4),  
    location_id char(4)  
);
```

```
create table employee8 (  
    employee_id char(4) primary key,  
    last_name varchar2(50) not null,  
    email varchar2(50),  
    salary number,
```

```
commission_pct number,  
hire_date date not null,  
department_id char(4),  
foreign key (department_id) references department(department_id)  
);
```

```
create table inventory8 (  
item_id char(4) primary key,  
item_name varchar2(50),  
quantity number,  
price number,  
supplier_id char(4),  
foreign key (supplier_id) references supplier(supplier_id),  
department_id char(4),  
foreign key (department_id) references department(department_id)  
);
```

```
create table supplier8 (  
supplier_id char(4) primary key,  
supplier_name varchar2(50),  
contact_name varchar2(50),  
phone varchar2(12)  
);
```

```
create table location8 (  
location_id char(4) primary key,  
department_name varchar2(50),  
country_id char(4)  
);
```

```
create table manager8 (  
manager_id char(4) primary key,  
manager_name varchar2(50)  
);
```

```
alter table department8  
add constraint manager_fk foreign key (manager_id) references  
manager8(manager_id);
```

```
alter table department8  
add constraint location_fk foreign key (location_id) references location8(location_id);
```

```
alter table employee8  
add constraint department_fk foreign key (department_id) references  
department8(department_id);
```

```
alter table inventory8  
add constraint supplier_fk foreign key (supplier_id) references supplier8(supplier_id);
```

```
alter table inventory8  
add constraint department_fk2 foreign key (department_id) references  
department8(department_id);
```

```
-- NOMOR 2
```

```
analyze table department8 compute statistics;  
analyze table employee8 compute statistics;  
analyze table inventory8 compute statistics;  
analyze table manager8 compute statistics;
```

analyze table location8 compute statistics;

analyze table supplier8 compute statistics;

-- NOMOR 3

INSERT ALL

INTO manager8 VALUES ('M001', 'Andi Wijaya')

INTO manager8 VALUES ('M002', 'Budi Santoso')

INTO manager8 VALUES ('M003', 'Citra Dewi')

INTO manager8 VALUES ('M004', 'Dedi Kurniawan')

INTO manager8 VALUES ('M005', 'Eka Putri')

INTO manager8 VALUES ('M006', 'Fajar Nugroho')

INTO manager8 VALUES ('M007', 'Gina Lestari')

INTO manager8 VALUES ('M008', 'Hadi Prasetyo')

INTO manager8 VALUES ('M009', 'Indah Sari')

INTO manager8 VALUES ('M010', 'Joko Susilo')

SELECT * FROM dual;

INSERT ALL

INTO location8 VALUES ('L001', 'Dept A', 'ID')

INTO location8 VALUES ('L002', 'Dept B', 'ID')

INTO location8 VALUES ('L003', 'Dept C', 'ID')

INTO location8 VALUES ('L004', 'Dept D', 'ID')

INTO location8 VALUES ('L005', 'Dept E', 'ID')

INTO location8 VALUES ('L006', 'Dept F', 'ID')

INTO location8 VALUES ('L007', 'Dept G', 'ID')

INTO location8 VALUES ('L008', 'Dept H', 'ID')

INTO location8 VALUES ('L009', 'Dept I', 'ID')

```
INTO location8 VALUES ('L010', 'Dept J', 'ID')
```

```
SELECT * FROM dual;
```

```
INSERT ALL
```

```
INTO department8 VALUES ('D001', 'HR', 'M001', 'L001')
```

```
INTO department8 VALUES ('D002', 'Finance', 'M002', 'L002')
```

```
INTO department8 VALUES ('D003', 'IT', 'M003', 'L003')
```

```
INTO department8 VALUES ('D004', 'Marketing', 'M004', 'L004')
```

```
INTO department8 VALUES ('D005', 'Sales', 'M005', 'L005')
```

```
INTO department8 VALUES ('D006', 'Logistik', 'M006', 'L006')
```

```
INTO department8 VALUES ('D007', 'Produksi', 'M007', 'L007')
```

```
INTO department8 VALUES ('D008', 'QA', 'M008', 'L008')
```

```
INTO department8 VALUES ('D009', 'Support', 'M009', 'L009')
```

```
INTO department8 VALUES ('D010', 'Legal', 'M010', 'L010')
```

```
SELECT * FROM dual;
```

```
INSERT ALL
```

```
INTO supplier8 VALUES ('S001', 'PT Maju Jaya', 'Rina', '0811111111')
```

```
INTO supplier8 VALUES ('S002', 'PT Sumber Rejeki', 'Budi', '0822222222')
```

```
INTO supplier8 VALUES ('S003', 'PT Makmur', 'Andi', '0833333333')
```

```
INTO supplier8 VALUES ('S004', 'PT Sentosa', 'Sari', '0844444444')
```

```
INTO supplier8 VALUES ('S005', 'PT Abadi', 'Dewi', '0855555555')
```

```
INTO supplier8 VALUES ('S006', 'PT Berkah', 'Joko', '0866666666')
```

```
INTO supplier8 VALUES ('S007', 'PT Mandiri', 'Lina', '0877777777')
```

```
INTO supplier8 VALUES ('S008', 'PT Sejahtera', 'Tono', '0888888888')
```

```
INTO supplier8 VALUES ('S009', 'PT Global', 'Nina', '0899999999')
```

```
INTO supplier8 VALUES ('S010', 'PT Nusantara', 'Rudi', '0810000000')
```

```
SELECT * FROM dual;
```

```
select * from supplier8;
```

```
INSERT ALL
```

```
INTO employee8 VALUES ('E001', 'Rahman', 'rahman@mail.com', 5000, 0.1, DATE  
'2020-01-01', 'D001')
```

```
INTO employee8 VALUES ('E002', 'Siti', 'siti@mail.com', 5500, 0.1, DATE  
'2020-02-01', 'D002')
```

```
INTO employee8 VALUES ('E003', 'Andi', 'andi@mail.com', 6000, 0.2, DATE  
'2020-03-01', 'D003')
```

```
INTO employee8 VALUES ('E004', 'Dewi', 'dewi@mail.com', 6500, 0.2, DATE  
'2020-04-01', 'D004')
```

```
INTO employee8 VALUES ('E005', 'Rudi', 'rudi@mail.com', 7000, 0.1, DATE  
'2020-05-01', 'D005')
```

```
INTO employee8 VALUES ('E006', 'Lina', 'lina@mail.com', 5200, 0.1, DATE  
'2020-06-01', 'D006')
```

```
INTO employee8 VALUES ('E007', 'Budi', 'budi@mail.com', 5800, 0.2, DATE  
'2020-07-01', 'D007')
```

```
INTO employee8 VALUES ('E008', 'Nina', 'nina@mail.com', 6200, 0.1, DATE  
'2020-08-01', 'D008')
```

```
INTO employee8 VALUES ('E009', 'Tono', 'tono@mail.com', 5400, 0.1, DATE  
'2020-09-01', 'D009')
```

```
INTO employee8 VALUES ('E010', 'Sari', 'sari@mail.com', 6100, 0.2, DATE  
'2020-10-01', 'D010')
```

```
SELECT * FROM dual;
```

```
INSERT ALL
```

```
INTO inventory8 VALUES ('I001', 'Laptop', 10, 10000, 'S001', 'D003')
```

```
INTO inventory8 VALUES ('I002', 'Mouse', 50, 200, 'S002', 'D003')
```

```
INTO inventory8 VALUES ('I003', 'Keyboard', 40, 300, 'S003', 'D003')
```

```
INTO inventory8 VALUES ('I004', 'Printer', 15, 2500, 'S004', 'D002')
```

```
INTO inventory8 VALUES ('I005', 'Scanner', 12, 2000, 'S005', 'D002')
```

```
INTO inventory8 VALUES ('I006', 'Chair', 30, 500, 'S006', 'D001')
```

```
INTO inventory8 VALUES ('I007', 'Table', 20, 800, 'S007', 'D001')
INTO inventory8 VALUES ('I008', 'Projector', 8, 3000, 'S008', 'D004')
INTO inventory8 VALUES ('I009', 'Whiteboard', 25, 400, 'S009', 'D004')
INTO inventory8 VALUES ('I010', 'AC Unit', 5, 5000, 'S010', 'D006')
SELECT * FROM dual;
```

```
select * from inventory8;
```

```
SELECT constraint_name, table_name
FROM user_constraints
WHERE table_name IN ('EMPLOYEE8','INVENTORY8');
```

```
ALTER TABLE employee8 DROP CONSTRAINT SYS_C008453;
ALTER TABLE employee8 DROP CONSTRAINT SYS_C008454;
ALTER TABLE employee8 DROP CONSTRAINT SYS_C008455;
ALTER TABLE employee8 DROP CONSTRAINT SYS_C008456;
```

```
ALTER TABLE inventory8 DROP CONSTRAINT SYS_C008457;
ALTER TABLE inventory8 DROP CONSTRAINT SYS_C008458;
ALTER TABLE inventory8 DROP CONSTRAINT SYS_C008459;
```

-- NOMOR 4

```
analyze table department8 compute statistics;
analyze table employee8 compute statistics;
analyze table inventory8 compute statistics;
analyze table manager8 compute statistics;
analyze table location8 compute statistics;
```



```
analyze table supplier8 compute statistics;
```

```
-- NOMOR 5
```

```
CREATE VIEW empuv80 AS  
SELECT  
    employee_id AS id_number,  
    last_name AS name,  
    salary,  
    department_id  
FROM employee8  
WHERE department_id = 'D008';
```

```
-- NOMOR 6
```

```
SELECT * FROM user_tab_columns  
WHERE table_name = 'EMPUV80';
```

```
-- NOMOR 7
```

```
CREATE INDEX idx_emp_dept  
ON employee8(department_id);
```